

Problem Set 1

Due: September 27, 2026, at 11:59 p.m.

1. [20 points] Consider a venture that succeeds with probability $1/2$. For each dollar invested, you earn a profit of \$1.60 if the venture succeeds and lose the invested dollar if it fails. You may invest repeatedly, and each time you invest exactly half of your current wealth. Starting with \$1,000,000:

- (i) **[5 points]** What is the effect of combining this investment opportunity with the strategy of always investing half of your current wealth?
- (ii) **[5 points]** Do you gain or lose money on average?
- (iii) **[10 points]** If you make 10,000 independent investments of this kind, what is your expected wealth after the final investment? What is the probability that your final wealth is less than the \$1,000,000 with which you started?

2. [10 points] A standard 52-card deck is well shuffled, and five cards are dealt without replacement. Find the probability of each of the following mutually exclusive poker-hand categories: royal flush; other straight flush; four of a kind; full house; flush (excluding straight flushes); straight (excluding straight flushes); three of a kind; two pairs; one pair; and high card only (none of the preceding categories). Show how you count the number of hands in each category, and verify that your counts sum to $\binom{52}{5}$.

3. [10 points] Let $\mathcal{F}$ be a σ -field on Ω . Prove that $\mathcal{F}$ is closed under intersections and set differences; that is, for any $A, B \in \mathcal{F}$, show that

$$A \cap B \in \mathcal{F} \quad \text{and} \quad A \setminus B \in \mathcal{F}.$$

4. [10 points] Suppose that we are given a measure space $(\Omega, \mathcal{F}, \mu)$. Show that whenever $A \subseteq B$, we have

$$\mu(A) \leq \mu(B).$$

Make explicit at each step where the defining properties of a measure and a σ -field are used.

5. [20 points] Let $\Omega = \{R, M, L\}$ and take the σ -field $\mathcal{F} = 2^\Omega$. Let P be a probability measure on $(\Omega, \mathcal{F})$ such that

$$P(\{R\}) = \frac{1}{2}, \quad P(\{M\}) = \frac{1}{4}, \quad P(\{L\}) = \frac{1}{4}.$$

Define random variables X and Y by, for every $\omega \in \Omega$,

$$X(\omega) = 2 \mathbf{1}\{\omega = R\} + 3 \mathbf{1}\{\omega \in \{M, L\}\},$$

$$Y(\omega) = 2 \mathbf{1}\{\omega \in \{M, L\}\} + 3 \mathbf{1}\{\omega = R\}.$$

- (i) **[10 points]** Find the cumulative distribution function of X and sketch its graph.
- (ii) **[10 points]** Find $\sigma(X)$, the σ -field generated by X .

6. [20 points] Suppose that we are given a measurable space $(\Omega, \mathcal{F})$.

- (i) **[10 points]** Suppose that $f : \Omega \rightarrow \mathbb{R}^d$ is measurable and $g : \mathbb{R}^d \rightarrow \mathbb{R}^r$ is Borel measurable; that is, for every $B \in \mathcal{B}(\mathbb{R}^r)$,

$$g^{-1}(B) \in \mathcal{B}(\mathbb{R}^d).$$

Show that $g \circ f$ is measurable.

- (ii) **[10 points]** Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable and let $f : \mathbb{R} \rightarrow \mathbb{R}$ be Borel measurable. Prove that the σ -field generated by X contains the σ -field generated by $f(X)$; that is, show that

$$\sigma(f(X)) \subseteq \sigma(X).$$

7. [10 points] Give an example of a probability space $(\Omega, \mathcal{F}, P)$ and two random variables X and Y such that X and Y have the same distribution but are not the same random variable.